

MWD Downlinking Simulation

Sending instructions down the hole three different ways

⚠ FOR UNDERSTANDING PURPOSES ONLY

Prasanaa Kartick T — Personal Learning Project

Flutter / Dart

RPM modulation

Mud flow

Wired pipe

Instruction tables

1. What It Is

A **downlink** is a message sent from the surface to the tool in the well, for example to change which measurements it records. This simulation teaches how that message is “spoken”. It is reached from Operations › LWD / MWD › MWD & DD › Downlinking and has three tabs, one for each method.

Tab	How the message travels
RPM Modulation (battery-powered MWD)	By changing the speed of the drill string in a set pattern
Mud Flow Modulation (computer controlled)	By changing the pump flow rate in a set pattern
Direct Downlink (wired pipe telemetry)	By typing commands that travel along cable built into the drill pipe

2. RPM Modulation

The upper half shows a **rotating drill-string circle** (the string tool phase) and a graph of RPM against time. The lower half has the controls.

1. Choose which setting to change: **Downhole Parameters**, **Gamma Ray (GR)**, **Resistivity** or **Power Management**.
2. An **instruction table** appears listing the exact sequence: for example INCREASE, hold for eight seconds, DECREASE, hold for nine seconds, and so on.
3. Press **Send Downlink**. A timer starts, and you press INCREASE or DECREASE at the right moments so the RPM swings between a low and a high value around a base speed.
4. Each step shows as selected, running or done. When every step is correct the screen shows **Successful** and the tool “receives” its new setting.

A fast-forward button speeds the timer up, and **Reset** clears the attempt.

3. Mud Flow Modulation

The same idea, but the signal is carried by the pump. The upper half shows a **rig diagram** and a **flow graph** in gallons per minute. The lower half repeats the parameter buttons and instruction table, and the learner raises and lowers the pump flow in the required pattern. A bottom bar reports the status of the downlink.

4. Direct Downlink

With wired pipe telemetry no pattern is needed: commands travel straight down the pipe. The screen shows a **wired pipe diagram** with information badges and an About panel, a set of **command buttons** on the left and a **command log** on the right that records each command with the time.

5. The Screen Parts Worth Noting

- **Shared pattern.** All three tabs have a diagram above and controls below, with a status bar and a Reset button in one place.
- **Hand-drawn graphics.** The rotating circle, the RPM graph and the flow graph are drawn directly by the app, including axes, arrow heads and tick labels, rather than using a chart library.
- **Tool response.** After a successful downlink the tool's new settings appear, such as formation and lithology readings for the gamma ray and resistivity parameters.